

Incremental Data-Driven Robust Crew Pairing Optimization

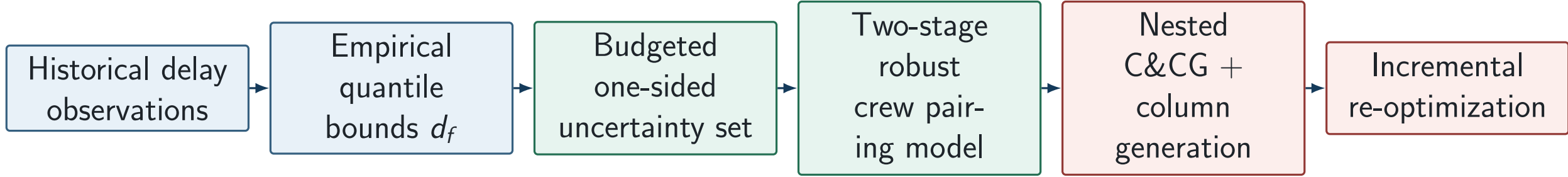
under Evolving Delay Uncertainty

Keywords
air crew pairing
robust optimization
C&CG and column generation

Research Question and Core Idea

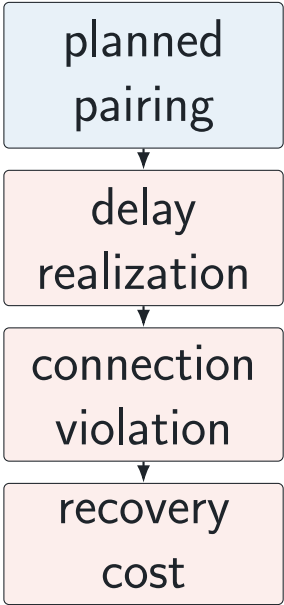
How can an airline build crew pairings that remain operationally reliable when delay data keep changing? The project combines **quantile-based delay bounds**, **budgeted uncertainty**, a **two-stage robust crew pairing model**, and **incremental re-optimization** so that new disruption information updates the model without rebuilding the entire optimization process from zero.

Poster Map: From Delay Data to an Updated Robust Pairing Plan



Operational Motivation

- Crew pairing assigns flight legs to crew duties while satisfying coverage, connection, rest, duty-time, hotel, and cost rules. Nominal pairings can be cheap but fragile.
- ▶ An upstream flight delay may break a crew connection.
 - ▶ Recovery actions include waiting, swapping, or cancellation.
 - ▶ Rare disruption days are exactly the cases that drive operational risk.
 - ▶ Early arrivals usually create no penalty, so the uncertainty is naturally one-sided.



Literature Positioning

Stream	Main idea	Use in this project
Robust optimization	Budgeted uncertainty controls the price of robustness [1].	Tunable conservatism
Two-stage robust optimization	C&CG adds worst-case scenarios iteratively [3].	Master and scenario cuts
Mixed-integer recourse	Nested C&CG handles integer recovery decisions [2].	Swap/cancel modeling
Transportation robustness	One-sided interval sets and budget parameters appear in service network design [4].	Delay-bound motivation

Research Gap

Existing work provides strong building blocks, but crew pairing under evolving disruption data needs the following combination:

- ▶ uncertainty bounds learned directly from historical observations;
- ▶ one-sided treatment of delay rather than symmetric distribution fitting;
- ▶ explicit mixed-integer recovery actions in the recourse stage;
- ▶ a warm-startable structure for updated data and new operational cuts.

Position: the project connects data-driven uncertainty construction with robust crew pairing and incremental solution updates.

Data-Driven Uncertainty Set

For each flight component f , historical delays $x_f^1, \dots, x_f^T$ define an empirical upper bound:

$$d_f = \text{Quantile}_\alpha(x_f^1, \dots, x_f^T), \quad 0 \leq \Delta_f \leq d_f.$$

The system does not assume all flights reach their worst delay simultaneously. A budget Γ allocates disruption intensity:

$$\tilde{\mathcal{U}}(\Gamma) = \left\{ \Delta : \Delta_f = \gamma_f d_f, 0 \leq \gamma_f \leq 1, \sum_{f \in F} \gamma_f \leq \Gamma \right\}.$$

Interpretation

- ▶ d_f controls the severity of each flight-level disruption.
- ▶ Γ controls network-wide conservatism.
- ▶ Quantiles reduce the dilution of rare disruptions by many nominal observations.

Experiment and Validation Blueprint

Inputs and instance design

- ▶ flight set F , feasible connection set A , and initial pairing pool $P^{(0)}$;
- ▶ empirical delay samples $x_f^1, \dots, x_f^T$ for each flight component;
- ▶ multiple instance scales to test how the nested C&CG and pricing layers behave as the pairing space grows.

Parameter analysis

- ▶ quantile level α : controls flight-level delay severity d_f ;
- ▶ uncertainty budget Γ : controls how many components can be simultaneously stressed;
- ▶ resource limits R^{swap} and R^{cancel} : represent operational recovery capacity.

Comparisons

- ▶ nominal crew pairing without robust recovery;
- ▶ robust model solved from scratch after data updates;
- ▶ incremental re-optimization that reuses old cuts, pairings, and scenarios;
- ▶ optional decomposition baselines, such as Benders-style cuts when applicable.

Reported metrics

- ▶ planned pairing cost and worst-case recovery cost;
- ▶ optimality gap $UB - LB$, number of C&CG cuts, and number of generated pairings;
- ▶ runtime, re-optimization speedup, and robustness price as Γ increases.

Expected interpretation. The experimental section should show whether additional conservatism buys meaningful disruption protection, whether integer recovery decisions change the structure of selected pairings, and whether incremental updates reduce computation when new delay observations or new operational restrictions arrive.

Takeaway

The project treats crew pairing as a living robust optimization problem: delay observations define plausible disruption bounds, the robust model protects against high-impact scenarios, and the algorithm updates the solution as uncertainty and operational constraints evolve.

First-Stage Crew Pairing Model

Let P be the generated feasible pairing pool and $a_{fp} = 1$ if pairing p covers flight f . The planning decision selects pairings before the delay scenario is known:

$$\min_x \sum_{p \in P} c_p x_p + \max_{\Delta \in \tilde{\mathcal{U}}(\Gamma)} R(x, \Delta)$$
$$\sum_{p \in P} a_{fp} x_p = 1 \quad \forall f \in F, \quad x_p \in \{0, 1\}.$$

Planning trade-off

- ▶ Cheap pairings may have tight buffers and high disruption exposure.
- ▶ Robust pairings spend more planned cost to reduce worst-case recovery cost.
- ▶ The model internalizes this trade-off in the max-recourse term.

Second-Stage Recourse

Given selected pairings and a disruption allocation, excess delay on connection (f, p) is:

$$z_{fp} \geq \gamma_f d_f - b_{fp}.$$

The airline then chooses between waiting, swapping, and cancellation:

$$R(x, \gamma) = \min_{(f,p) \in A} x_p (c_p^{\text{wait}} w_{fp} + c_{fp}^{\text{swap}} y_{fp}^{\text{swap}} + c_{fp}^{\text{cancel}} y_{fp}^{\text{cancel}})$$
$$\text{s.t. } w_{fp} \geq z_{fp} - M(y_{fp}^{\text{swap}} + y_{fp}^{\text{cancel}}),$$
$$y_{fp}^{\text{swap}} + y_{fp}^{\text{cancel}} \leq 1,$$
$$\sum_p x_p y_{fp}^{\text{swap}} \leq R^{\text{swap}}, \quad \sum_p x_p y_{fp}^{\text{cancel}} \leq R^{\text{cancel}}.$$

Why this matters

The recourse is not just a continuous delay penalty. Swap and cancel decisions are binary, so direct strong-duality conversion of the full recourse problem is not valid.

Worst-Case Separation

For a fixed first-stage solution $\bar{x}$, the model seeks the delay allocation that creates the largest optimized recovery cost:

$$Q(\bar{x}) = \max_{\gamma \in \tilde{\mathcal{U}}(\Gamma)} \min_{w, x, y^{\text{swap}}, y^{\text{cancel}}} \{\text{recovery cost}\}.$$

Because the recourse contains integer actions, the project adopts a nested C&CG idea:

- ▶ fix a candidate disruption allocation;
- ▶ solve the mixed-integer recourse problem;
- ▶ add the corresponding recovery pattern or scenario cut;
- ▶ repeat until no violated worst-case pattern remains.

Restricted Master with Scenario Cuts

At outer iteration k , the master problem keeps a current pairing pool $P^{(k)}$ and accumulated worst-case scenarios:

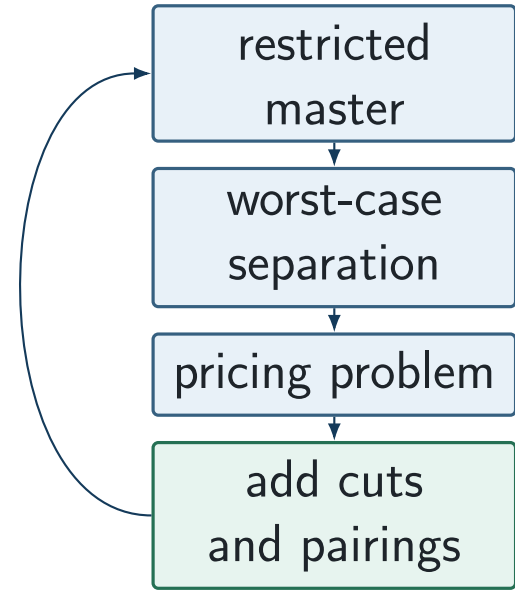
$$\min_{x, \theta} \sum_{p \in P^{(k)}} c_p x_p + \theta$$
$$\theta \geq \sum_{(f,p) \in A} x_p (c_p^{\text{wait}} w_{fp}^s + c_{fp}^{\text{swap}} y_{fp}^{\text{swap},s} + c_{fp}^{\text{cancel}} y_{fp}^{\text{cancel},s}), \quad s = 1, \dots, k.$$

Each new scenario tightens the lower bound on robust recovery cost.

Column Generation Layer

Crew pairing has too many feasible pairings to enumerate. Column generation searches for pairings with negative reduced cost under the current dual multipliers.

$$rc(p) = c_p - \sum_{f \in F} a_{fp} \kappa_f + \sum_s \text{recourse-adjusted dual contribution}.$$



Termination

The algorithm stops when no negative-reduced-cost pairing remains and no violated robust scenario is generated, or when the optimality gap satisfies $UB - LB \leq \varepsilon$.

Incremental Re-Optimization

Let $\mathcal{U}(\beta)$ be the uncertainty set parameterized by delay bounds β . If new data update these bounds, monotonicity gives two useful cases:

$$\beta^{\text{new}} \leq \beta^{\text{old}} \Rightarrow \mathcal{U}(\beta^{\text{new}}) \subseteq \mathcal{U}(\beta^{\text{old}}) \Rightarrow X(\beta^{\text{old}}) \subseteq X(\beta^{\text{new}}).$$

$$\beta^{\text{new}} \geq \beta^{\text{old}} \Rightarrow \mathcal{U}(\beta^{\text{new}}) \supseteq \mathcal{U}(\beta^{\text{old}}) \Rightarrow X(\beta^{\text{new}}) \subseteq X(\beta^{\text{old}}).$$

- ▶ If uncertainty shrinks, the old robust solution remains feasible and conservative.
- ▶ If uncertainty expands, keep old cuts and add only new violated scenario cuts.
- ▶ Operational cuts, such as restricted swap availability, can be inserted without restarting the full model.

Numerical Study Plan

Question	Evaluation design
Robustness price	Vary Γ and quantile level α ; compare planned cost and worst-case recovery cost.
Solution quality	Track lower bound, upper bound, optimality gap, number of cuts, and number of generated pairings.
Re-optimization value	Compare incremental updating against solving the enlarged problem from scratch.
Operational policy	Test swap and cancellation limits, including new cuts when an airline changes recovery availability.

Expected Contributions

- ▶ A one-sided empirical quantile uncertainty set for disruption-driven airline delay risk.
- ▶ A two-stage robust crew pairing formulation with explicit waiting, swap, and cancellation recourse.
- ▶ A nested C&CG and column-generation solution framework for large pairing spaces.
- ▶ An incremental update mechanism that reuses previous scenarios, cuts, and pairing columns as uncertainty evolves.

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